

The zero-spacing limit for any plaquette action and any dimension

Result, 2026-09-27. The series/parallel note worked with the heat-kernel action. Here the question is how the $a \rightarrow 0$ limit depends on the choice of plaquette action, and on the dimension.

1. **Two semigroups.** A series move convolves plaquette weights, so it multiplies their character coefficients; the heat kernel is the family closed under it. A parallel move multiplies weights pointwise, so it adds their logarithms; exponential families $e^{-\beta V}$ (Wilson, Villain-type) are closed under it. In the small-field regime the two agree: for $U(1)$ the product of two heat kernels is exactly a heat kernel at the parallel heat time $st/(s+t)$ plus vortex terms of relative size $e^{-2\pi^2/(s+t)}$ (Proposition 1).
2. **Two dimensions, any action** (Theorem 2). The possible continuum limits of refining a two-dimensional lattice gauge theory with any symmetric class-function plaquette weight are exactly the conjugation-invariant Lévy exponents $\psi(R) = \frac{\sigma^2}{2}C_2(R) + \int(1 - \operatorname{Re} \chi_R/d_R) d\nu$, with $\psi(R) = \lim a^{-2}(1 - \hat{c}_R(a))$. Yang–Mills is the Gaussian member, $\nu = 0$, selected by a Lindeberg condition on the action. A weight with an atom at a centre element produces a centre-vortex gas, and the circle spectrum becomes $E_R - E_0 = \hbar c L \psi(R)$.
3. **Any dimension: an error budget per physical volume.** One refinement step leaves three kinds of error. The perturbative part is of relative order t_n , the plaquette heat time at step n . The large-field part has density $a_n^{-D} e^{-c/t_n}$ per physical volume, with c the minimal large-field action of the chosen lattice action in units of the inverse heat time. The jump part is the action's Lévy content. With $t \propto \lambda_D a^{4-D}$:
 - $D < 4$: all three are summable at fixed λ_D once the jump part vanishes, which is why the lattice gap of compact $U(1)$ disappears at fixed coupling (Gross, Theorem 5 of the series/parallel note);
 - $D = 4$: the perturbative part resums into the running coupling, and the large-field part decays only as a power of a , $(a\Lambda)^{2b_0c}$, so it dies iff $2b_0c > 4$, that is $c > 96\pi^2/(11N)$ for pure $SU(N)$: $c > 43.1$ for $SU(2)$, $c > 28.7$ for $SU(3)$, that is $12/(11N)$ of the instanton action. Instantons ($c = 8\pi^2 \approx 79$) pass for every N ; for lattice dislocations this is the known 6/11 criterion of Pugh–Teper and Gökeler et al. (1989);
 - $D > 4$: t grows as $a \rightarrow 0$ at fixed λ_D , so no small-field regime exists at short distances.

So, in the ultraviolet, the limit depends on the lattice action only through its second moment (the coupling, and in $D = 4$ the ratio of Λ parameters), its minimal large-field action and its jump content. A gap, being an infrared property, is then the same for every action that meets the three conditions, up to the Λ ratio. The ingredients are established (character expansions, Poisson summation, Hunt's classification of convolution semigroups, the one-loop running); the organization by the two semigroups and the error budget is the contribution, with no novelty claimed for the parts.

1. Series and parallel act on different coordinates of the weight

Let w be a class-function probability density on a compact connected Lie group G (Haar measure normalized), symmetric, $w(g^{-1}) = w(g)$, with character coefficients $\hat{c}_R = d_R^{-1} \int w \overline{\chi_R} dg \in [-1, 1]$, so that $w = \sum_R d_R \hat{c}_R \chi_R$. Group metric as in the series/parallel note §1 ($-\Delta \chi_R = C_2(R) \chi_R$).

- **Series.** Integrating out an edge shared by two faces convolves their weights, and $(w_1 * w_2)_R = \hat{c}_R(w_1) \hat{c}_R(w_2)$. The heat kernels k_t , $\hat{c}_R = e^{-tC_2(R)/2}$, form the one-parameter family closed under this product with t additive.

- **Parallel.** Two faces with the same holonomy combine by the pointwise product $w_1 w_2$, and $\log(w_1 w_2) = \log w_1 + \log w_2$. The families $w_\beta \propto e^{-\beta V}$ with fixed V are closed under this operation with β additive; Wilson's $V = 1 - \text{Re tr } U/N$ is one.

Proposition 1 (parallel product of $U(1)$ heat kernels). With $k_s(\theta) = \sum_{n \in \mathbb{Z}} e^{-sn^2/2} e^{in\theta}$ and $\tau = st/(s+t)$,

$$k_s(\theta) k_t(\theta) = \frac{1}{\sqrt{2\pi(s+t)}} \sum_{m \in \mathbb{Z}} e^{-2\pi^2 m^2/(s+t)} k_\tau\left(\theta + \frac{2\pi ms}{s+t}\right).$$

Proof. Poisson summation gives $k_s(\theta) = \sum_w G_s(\theta + 2\pi w)$ with $G_s(x) = (2\pi s)^{-1/2} e^{-x^2/(2s)}$. For Gaussians, $G_s(x) G_t(y) = G_{s+t}(x-y) G_\tau((tx+sy)/(s+t))$. Put $x = \theta + 2\pi w$, $y = \theta + 2\pi w'$ and $m = w' - w$; the sum over w at fixed m is a wrapped Gaussian of variance τ centred at $-2\pi ms/(s+t)$, which is k_τ at the shifted argument by Poisson summation again. $\square$

The $m = 0$ term is the heat kernel at the parallel heat time, the conductance rule of the series/parallel note §3. The terms $m \neq 0$ are vortices, of relative weight $e^{-2\pi^2 m^2/(s+t)}$. So the heat kernel is closed under series moves exactly and under parallel moves up to $e^{-c/t}$ corrections; exponential families have the opposite property. Both reduce to the Gaussian in the small-field regime, where they agree.

A limit of validity for heat-kernel anisotropy. Equation (1) of the series/parallel note matches heat times to the continuum action in the small-field regime. Refining one direction alone, for instance Euclidean time at fixed spatial lattice, sends the transverse heat times to infinity. Heat-kernel weights then become flat up to corrections $e^{-tC_2/2}$, whereas the Kogut–Susskind limit needs a magnetic weight $e^{-a_0 V}$ with a_0 the time step. The Hamiltonian limit is reached within the exponential family. The directional halvings of an isotropic cycle change heat times only by factors of two, so the propositions of that note are unaffected.

2. Two dimensions: the continuum limits of every plaquette action

In two dimensions every refinement is a series move (the series/parallel note, §2). A face of area A tiled by A/a^2 plaquettes of weight w_a has, after integrating the interior edges, the weight whose coefficients are $\hat{c}_R(a)^{A/a^2}$.

Theorem 2. Let w_a be symmetric class-function plaquette weights on the lattice of spacing a , with $\hat{c}_R(a) \rightarrow 1$ for every R . Suppose that for every R the limit

$$\psi(R) = \lim_{a \rightarrow 0} \frac{1 - \hat{c}_R(a)}{a^2} = \lim_{a \rightarrow 0} a^{-2} \int_G \left(1 - \frac{\text{Re } \chi_R(g)}{d_R}\right) w_a(g) dg$$

exists. Then every face of area A has the limiting coefficients $e^{-A\psi(R)}$, and ψ has the form

$$\psi(R) = \frac{\sigma^2}{2} C_2(R) + \int_{G \setminus \{e\}} \left(1 - \frac{\text{Re } \chi_R(g)}{d_R}\right) \nu(dg),$$

with $\sigma^2 \geq 0$ and ν a conjugation-invariant symmetric measure satisfying

$$\int_G \min(1, d(e, g)^2) \nu(dg) < \infty.$$

The continuum theory is Yang–Mills with $\lambda_2 = \sigma^2$ exactly when $\nu = 0$, and $\nu = 0$ holds when, for every $\varepsilon > 0$, $a^{-2} \int_{d(e, g) > \varepsilon} w_a dg \rightarrow 0$.

Proof. Since $\hat{c}_R(a) \rightarrow 1$, $(A/a^2) \log \hat{c}_R(a) = -(A/a^2)(1 - \hat{c}_R(a))(1 + o(1)) \rightarrow -A\psi(R)$. Split the integral defining ψ at $d(e, g) = \varepsilon$. Near the identity, $1 - \text{Re } \chi_R(e^X)/d_R = C_2(R)|X|^2/(2 \dim G) + O(|X|^3)$ after averaging over the directions of X , which conjugation invariance of w_a permits; this part converges to $\frac{\sigma^2}{2} C_2(R)$ with

$$\sigma^2 = \lim_{\varepsilon \rightarrow 0} \lim_{a \rightarrow 0} \frac{1}{a^2 \dim G} \int_{d(e,g) < \varepsilon} |X|^2 w_a(g) dg, \quad g = e^X,$$

along a subsequence if needed. Away from the identity, the measures $a^{-2} w_a dg$ restricted to $\{d \geq \varepsilon\}$ have bounded mass (a finite sum of the functions $1 - \operatorname{Re} \chi_R/d_R$ over representations separating the points of G is continuous and positive off e , hence bounded below there), so a subsequence converges weakly to ν on that set, conjugation invariant and symmetric. The integrability of ν near e follows from the bound on the near part. The form of ψ agrees with Hunt's classification of convolution semigroups on Lie groups, restricted to central ones (Hunt 1956, metadata). The Lindeberg condition sets the far part to zero. $\square$

The limiting objects are Lévy's two-dimensional Markovian holonomy fields, which he constructs from Lévy processes on G (Lévy, Astérisque 329, metadata). Theorem 2 adds only the lattice-side statement: which plaquette actions lead to which member.

Examples. The Wilson action $w_a \propto e^{\beta_a \operatorname{Re} \operatorname{tr} U}$ with $\beta_a \propto 1/(\lambda_2 a^2)$ concentrates like a Gaussian of width $\sqrt{\lambda_2} a$, satisfies the Lindeberg condition, and gives Yang–Mills. A weight carrying an explicit centre-vortex fugacity, $w_a = (1 - \kappa a^2) k_{\lambda_2 a^2} + \frac{\kappa a^2}{2} [k_{\lambda_2 a^2}(z \cdot) + k_{\lambda_2 a^2}(z^{-1} \cdot)]$ with z a generator of the centre $\mathbb{Z}_N$ of $SU(N)$, gives $\nu = \frac{\kappa}{2}(\delta_z + \delta_{z^{-1}})$ and

$$\psi(R) = \frac{\lambda_2}{2} C_2(R) + \kappa \left(1 - \cos \frac{2\pi k(R)}{N}\right),$$

with $k(R)$ the N -ality. On a spatial circle of circumference L the refinement note's eq. (10) becomes $E_R - E_0 = \hbar c L \psi(R)$: the vortex gas adds an N -ality-dependent string tension. The limit is action-dependent exactly through (σ^2, ν) .

3. Any dimension: the error budget of one step

Let t_n be the plaquette heat time at refinement step n , which by eq. (1) of the series/parallel note scales as $\lambda_D a_n^{4-D}$ at fixed physical coupling, and let L be a fixed physical box side. One directional step leaves three kinds of error.

- **Perturbative.** Coupling shifts and irrelevant operators of relative size $O(t_n)$ (Hypothesis P(α) and Proposition 7 of the series/parallel note).
- **Large-field.** Configurations far from the identity at the lattice scale, with action at least c/t_n , of density per physical volume about $a_n^{-D} e^{-c/t_n}$; here c depends on the lattice action (for heat-kernel $U(1)$ it is the vortex constant of Theorem 5 there).
- **Jump.** The action's Lévy content, the part of ν in Theorem 2.

The continuum limit discards the lattice structure when the three are summable over steps in a fixed physical box.

D	t_n , fixed λ_D	Perturbative	Large-field, $\sum (L/a_n)^D e^{-c/t_n}$
2	$\lambda_2 a_n^2 \rightarrow 0$	summable	summable; limit set by (σ^2, ν)
3	$\lambda_3 a_n \rightarrow 0$	summable	summable faster than any power
4	$g^2(a_n) \rightarrow 0$, logarithmically	resums into running	$\propto (a_n \Lambda)^{2b_0 c - 4}$, needs $2b_0 c > 4$
> 4	$\lambda_D a_n^{4-D} \rightarrow \infty$	no small parameter	no small-field regime

The perturbative column, with its sources. For $D < 4$, t_n falls geometrically along the dyadic sequence, so coupling shifts of relative order t_n sum to a finite renormalization: superrenormalizability. For three-dimensional lattice Yang–Mills, Balaban's ultraviolet stability is

the constructive form of this statement in his block-averaging scheme (Balaban 1985, metadata). For $D = 4$ the heat time is the coupling itself, and each isotropic step shifts g^{-2} by the same amount, $-2b_0 \log 2$ at one loop, with $b_0 = 11N/(48\pi^2)$ for pure $SU(N)$ (Gross and Wilczek 1973; Politzer 1973; metadata). These shifts do not sum; they are the running coupling. Different lattice actions share the first two coefficients of the running and differ by a finite factor in Λ , computed at one loop for the Wilson action by Hasenfratz and Hasenfratz (1980) (abstract as indexed) and Dashen and Gross (1981) (metadata); for pure $SU(3)$ with the Wilson action the standard value is $\Lambda_{\overline{\text{MS}}}/\Lambda_{\text{lat}} \approx 28.81$ (Capitani 2003, review, as indexed). The heat-kernel action as a lattice action is treated by Menotti and Onofri (1981) (metadata). For $D > 4$, t_n grows as $a \rightarrow 0$ at fixed λ_D ; lattice Monte Carlo finds a phase transition between the confining strong-coupling regime and a weak-coupling spin-wave phase for $SU(2)$ in five dimensions, and none in four (Creutz 1979, abstract). A continuum limit there would need a non-trivial ultraviolet fixed point, and none is established.

The four-dimensional threshold. With $g^{-2}(a) = 2b_0 \log(1/(a\Lambda))$, the convention of the refinement note §6, $e^{-c/g^2(a)} = (a\Lambda)^{2b_0 c}$. For pure $SU(N)$, $b_0 = 11N/(48\pi^2)$, so the large-field density per physical volume vanishes iff

$$c > \frac{2}{b_0} = \frac{96\pi^2}{11N} : \quad c > 43.1 \text{ } (SU(2)), \quad c > 28.7 \text{ } (SU(3)).$$

A continuum instanton has $c = 8\pi^2 \approx 79$, giving the density $(a\Lambda)^{11N/3} a^{-4}$ of small instantons, which vanishes for every $N \geq 2$ because $11N/3 > 4$. Lattice actions admit dislocations, small configurations of topological charge whose action lies below $8\pi^2$. In units of the instanton action the threshold reads $c/(8\pi^2) > 12/(11N)$, which is $6/11$ for $SU(2)$: this is the known dislocation criterion, found for the topological susceptibility by Pugh and Teper (1989) (abstract: with the plaquette action the $SU(2)$ susceptibility diverges in the continuum limit) and Gökeler, Kronfeld, Laursen, Schierholz and Wiese (1989) (metadata), and stated in the form “smaller than $6/11$ (for $SU(2)$) times the continuum value of a one-instanton configuration” by DeGrand, Hasenfratz and Zhu (1996) (passage). Admissibility conditions on lattice fields (Lüscher 1982, metadata) remove dislocations outright. The error budget places that criterion in a general frame: the same inequality decides, for any lattice action and any large-field configuration, whether its density per physical volume survives refinement. The thresholds carry the higher-order corrections of the running.

What this says about the mass gap. In $D = 3$ at fixed λ_3 every plaquette action with vanishing jump content has an ultraviolet limit governed by the same small-field analysis; a gap $C_3 \hbar c \lambda_3$ is then a property of the infrared and, if it exists, the same for all such actions. In $D = 4$ the same holds for actions whose large-field constant exceeds $2/b_0$, with gap $C_4 \hbar c \Lambda$ and a Λ parameter that depends on the action by a computable finite factor (the perturbative column above). The universality of the gap across actions is thereby reduced to the three ultraviolet conditions, and the gap itself remains the infrared obligation of the mass-gap map.

4. Consequence for STATE

Item 1 of STATE gains its action and dimension dependence. The next steps on this line are to state Hypothesis P(α) for a general symmetric class-function action (second moment, large-field constant and jump content as its data), and to connect the four-dimensional large-field threshold to the H1 blocking hypothesis of the conditional theorem. For the joint paper, Theorem 2 gives the two-dimensional row of the comparison for every action at once, and the table gives the other rows.